ETHAN KIM

CONTACT INFORMATION

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LinkedIn GitHub Portfolio

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EDUCATION

University of North Carolina at Chapel Hill
Bachelor of Science Computer Science
& Minor in Statistics and Operations
Expected 2029
Associates in Arts from Guilford Technical Community College
Relevant Courses
Fall 2026:
COMP 210 - Analysis of Data Structures
COMP 283 - Discrete Structures
STOR 305 - Intro Decision Analytics
MATH 233 - Multi Vari Calc 1 (Calc 3)
APPL 101 - Intro to Engineering
Spring 2027:
COMP 211 - Systems Fundamentals
COMP 301 - Foundations of Programming
Comp 455 - Models of Languages and Computation
MATH 347 - Linear Algebra for Applications
STOR 235 - Introduction to Probability

SKILLS

Languages
- Java
- Python
- JavaScript
- TypeScript
- HTML
- CSS
- Lua

Frameworks
- React
- Next.js
- FastAPI
- Flask
- Websockets
- SocketIO
- OpenCV
- YOLO

Robotics
- Kalman Filters for Sensor Fusion
- Control Systems
- Mapping
- Motion Planning
- Inverse Kinematics
- Computer Vision

Hardware
- Jetson Orin Nano & other Linux devices
- Arduino
- Realsense Depth Cameras
- GPS & RTK GPS
- Radio
- Lidar

CERTFICATIONS

Java Programming Certification from Guilford Technical Community College

EXPERIENCE AND PROJECTS

North Carolina Agriculture and Technical State University
CLAEM Internship - Intern (June - August 2026)
Helped PHD students create and implement **software** for **Outdoor Agricultural mobile robots** with **sensor fusion** for **localization**, **mapping**, **obstacle avoidance** and a **web based dashboard** with telemetry, mapping, camera perception and robot control.

- Developed **autonomous navigation** software for mobile robots using **LiDAR**, **depth cameras**, and **RTK GPS**.
- Designed **Kalman filter** sensor fusion integrating RTK GPS, **IMU**, and **wheeled odometry** for **outdoor localization**.
- **Combined LiDar** and **Depth Cameras perception** with localization to create a **Dynamic Obstacle Map**.
- Implemented **Dynamic Window Approach** local path planning for **obstacle avoidance** and navigation.
- Built a **Web interface** for robot telemetry and control with a python fastAPI backend, uvicorn server, a React TSX frontend, and with Websockets as the communication protocol.
- Trained and integrated **YOLO object detection** models for robotic arm use.
- Developed **inverse kinematics algorithms** for robotic arm positioning and movement.

Undertale Fighting Game Creator and Programmer
(March 2022 - Present)

- Created, designed and **programmed** a multiplayer Roblox game with over **6 million visits** and over **2 million unique players**
- Designed gameplay, visual effects, movement, and interfaces.
- Created and managed an online community supporting hundreds of players

First Tech Challenge Team 10195 Captain and Lead Programmer (May 2024 - June 2026)

- Led software development for a competitive First Tech Challenge Robot.
- Implemented **PID Control** for Motor Control based on Encoder and Camera data.
- Created a **Kalman Filters** for Sensor fusion between **Wheeled Odometry, IMU, and April Tags**.
- As lead programmer created robust autonomous routines and achieved North Carolina **Autonomous** Offensive Power Rating **State Record** in 2024 - 2025.
- As captain led the team to **win their first competition** since 2017 in the 2025 - 2026 season.